

CarbonLedger

Decision Table and Test Set

Complete decision table, derived tests and the reproducibility regression test

CL-DTT-001 · VERSION 1.0 · DRAFT FOR REVIEW

48

combinations

12

condensed rules

79

test cases

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Summary & Contents

Purpose of this document

Builds the 48-combination decision table for data source, scope, supplier declaration and period state, derives the test set from it, and designs the byte-for-byte regression test that recomputes a historical report.

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Document control

Revision history

Version	Date	Description	Author (role)
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Approvals

Role	Signature	Date
Sponsor (Manufacturer, Head of Sustainability)		
Reporting Lead (Manufacturer)		
Sole Engineer (all project roles)		
Reviewer (Faculty)		

Related documents

- CL-SRS-001 Software Requirements Specification (v1.0)
- CL-DTT-001 Decision Table and Test Set (v1.0) · this document
- CL-DSG-001 Design Pack (v1.0)
- CL-CMP-001 Configuration Management Plan (v1.0)
- CL-DQP-001 Data Quality Plan (v1.0)
- CL-PPL-001 Project Plan (v1.0)

Notice. This document is controlled; printed copies are uncontrolled. Figures stated as project data (data quality tiers of ±2%, ±8% and ±25%, nine plants, three scopes) come from the case study. Numbers in worked examples and test fixtures are synthetic, are labelled as such, and are not emission factors published by any authority.

1. Purpose and approach

This document turns the rules in the SRS (CL-SRS-001) into a **decision table**, derives a **test set** from it, and designs the **reproducibility regression test** that recomputes a historical report and compares it byte for byte. It contains 79 test cases.

Techniques used, and why each:

Technique	Applied to	Reason
Decision table testing	Data source x scope x declaration x period closed	The behaviour depends on a combination of four conditions; a table shows every combination and exposes gaps and overlaps.
Boundary value analysis	Activity dates at period ends, recorded_at at closed_at , month-crossing bills	Errors in date logic sit on boundaries.
Equivalence partitioning	Tiers, scopes, roles	One representative per class, plus invariants where a condition must make no difference.
Regression (golden master)	A real historical report	The core requirement is that history does not change.
Mutation check	The regression test itself	A test that cannot fail proves nothing.

2 Conditions and actions

Conditions (inputs given by the case study):

Condition	Values
C1 Data source	metered, invoiced, estimated, missing
C2 Scope	1, 2, 3
C3 Supplier declaration present	N, Y
C4 Period closed (at the time the data arrives or is evaluated)	N, Y

Total combinations: $4 \times 3 \times 2 \times 2 = 48$.

Actions (outcomes):

Code	Action	Meaning
A1	ACCEPT_METERED	Include in this period; tier metered, $\pm 2\%$.
A2	ACCEPT_INVOICED	Include in this period; tier invoiced, $\pm 8\%$.
A3	ACCEPT_ESTIMATED	Include in this period; tier estimated, $\pm 25\%$; flagged.
A4	USE_DECLARATION	Use the supplier's declared value; tier estimated, $\pm 25\%$; flag DECLARED (unverified).
A5	RAISE_OPEN_ITEM	Record an Open Item. At close it is gap-filled (estimated, $\pm 25\%$, flag GAP_FILLED) if a basis exists; otherwise close is blocked.
A6	LATE_ADJ_METERED	Late record: store, post an adjustment to the next open period; tier metered. Closed report unchanged.
A7	LATE_ADJ_INVOICED	As A6; tier invoiced.
A8	LATE_ADJ_ESTIMATED	As A6; tier estimated.

2. Conditions and actions

Code	Action	Meaning
A9	LATE_ADJ_DECLARED	As A6; declaration arrived late; tier estimated; flag DECLARED.
A10	CLOSED_STANDS_WATCH	Closed report keeps its gap-filled value; item stays on the Late-Data Watchlist; nothing recalculated, nothing posted.

For A6 to A9 the adjustment amount follows FR-LATE-04 (full emission, or the difference to the amount most recently reported).

Reading the awkward cases. A closed period can never contain an unresolved missing item, because close is refused until every Open Item is received or gap-filled (FR-PC-02). So "missing after close" means: the item was gap-filled at close and the real data has still not come. That is rule R11 or R12 (stays on the watchlist). If a supplier declaration arrives for it later, that is R10. A supplier declaration is only meaningful for Scope 3; for Scope 1 and 2 it is ignored (don't-care), which is tested explicitly.

3 Decision table (condensed, 12 rules)

A dash means "any value" (don't-care).

Rule	Period closed	Data source	Scope	Declaration	Action	Tier and uncertainty
R1	N	metered	any	-	A1 ACCEPT_METERED	METERED ±2%
R2	N	invoiced	any	-	A2 ACCEPT_INVOICED	INVOICED ±8%
R3	N	estimated	any	-	A3 ACCEPT_ESTIMATED	ESTIMATED ±25%
R4	N	missing	3	Y	A4 USE_DECLARATION	ESTIMATED ±25%, flag DECLARED
R5	N	missing	1 or 2	-	A5 RAISE_OPEN_ITEM	none yet; ESTIMATED ±25% if gap-filled at close
R6	N	missing	3	N	A5 RAISE_OPEN_ITEM	none yet; ESTIMATED ±25% if gap-filled at close
R7	Y	metered	any	-	A6 LATE_ADJ_METERED	METERED ±2%
R8	Y	invoiced	any	-	A7 LATE_ADJ_INVOICED	INVOICED ±8%
R9	Y	estimated	any	-	A8 LATE_ADJ_ESTIMATED	ESTIMATED ±25%
R10	Y	missing	3	Y	A9 LATE_ADJ_DECLARED	ESTIMATED ±25%, flag DECLARED
R11	Y	missing	1 or 2	-	A10 CLOSED_STANDS_WATCH	none; closed report keeps its gap-filled value
R12	Y	missing	3	N	A10 CLOSED_STANDS_WATCH	none; closed report keeps its gap-filled value

4 Completeness and consistency check

The full expansion (section 5) was generated by a program that, for every one of the 48 combinations, finds the rules whose conditions match. The program asserts that **exactly one rule matches each combination**.

Check	Result
Combinations covered	48 of 48

4. Completeness and consistency check

Check	Result
Combinations matching no rule (gaps)	0
Combinations matching more than one rule (conflicts)	0
Rules with don't-cares that expand to several combinations	R1, R2, R3, R7, R8, R9 (6 each); R5, R11 (4 each)
Rules matching exactly one combination	R4, R6, R10, R12

5

Full expansion (all 48 combinations)

#	Closed	Source	Scope	Decl	Rule	Action
1	N	metered	1	N	R1	A1 ACCEPT_METERED
2	N	metered	1	Y	R1	A1 ACCEPT_METERED
3	N	metered	2	N	R1	A1 ACCEPT_METERED
4	N	metered	2	Y	R1	A1 ACCEPT_METERED
5	N	metered	3	N	R1	A1 ACCEPT_METERED
6	N	metered	3	Y	R1	A1 ACCEPT_METERED
7	N	invoiced	1	N	R2	A2 ACCEPT_INVOICED
8	N	invoiced	1	Y	R2	A2 ACCEPT_INVOICED
9	N	invoiced	2	N	R2	A2 ACCEPT_INVOICED
10	N	invoiced	2	Y	R2	A2 ACCEPT_INVOICED
11	N	invoiced	3	N	R2	A2 ACCEPT_INVOICED
12	N	invoiced	3	Y	R2	A2 ACCEPT_INVOICED
13	N	estimated	1	N	R3	A3 ACCEPT_ESTIMATED
14	N	estimated	1	Y	R3	A3 ACCEPT_ESTIMATED
15	N	estimated	2	N	R3	A3 ACCEPT_ESTIMATED
16	N	estimated	2	Y	R3	A3 ACCEPT_ESTIMATED
17	N	estimated	3	N	R3	A3 ACCEPT_ESTIMATED
18	N	estimated	3	Y	R3	A3 ACCEPT_ESTIMATED
19	N	missing	1	N	R5	A5 RAISE_OPEN_ITEM
20	N	missing	1	Y	R5	A5 RAISE_OPEN_ITEM
21	N	missing	2	N	R5	A5 RAISE_OPEN_ITEM
22	N	missing	2	Y	R5	A5 RAISE_OPEN_ITEM
23	N	missing	3	N	R6	A5 RAISE_OPEN_ITEM
24	N	missing	3	Y	R4	A4 USE_DECLARATION
25	Y	metered	1	N	R7	A6 LATE_ADJ_METERED
26	Y	metered	1	Y	R7	A6 LATE_ADJ_METERED
27	Y	metered	2	N	R7	A6 LATE_ADJ_METERED
28	Y	metered	2	Y	R7	A6 LATE_ADJ_METERED
29	Y	metered	3	N	R7	A6 LATE_ADJ_METERED
30	Y	metered	3	Y	R7	A6 LATE_ADJ_METERED
31	Y	invoiced	1	N	R8	A7 LATE_ADJ_INVOICED
32	Y	invoiced	1	Y	R8	A7 LATE_ADJ_INVOICED
33	Y	invoiced	2	N	R8	A7 LATE_ADJ_INVOICED
34	Y	invoiced	2	Y	R8	A7 LATE_ADJ_INVOICED

5. Full expansion (all 48 combinations)

#	Closed	Source	Scope	Decl	Rule	Action
35	Y	invoiced	3	N	R8	A7 LATE_ADJ_INVOICED
36	Y	invoiced	3	Y	R8	A7 LATE_ADJ_INVOICED
37	Y	estimated	1	N	R9	A8 LATE_ADJ_ESTIMATED
38	Y	estimated	1	Y	R9	A8 LATE_ADJ_ESTIMATED
39	Y	estimated	2	N	R9	A8 LATE_ADJ_ESTIMATED
40	Y	estimated	2	Y	R9	A8 LATE_ADJ_ESTIMATED
41	Y	estimated	3	N	R9	A8 LATE_ADJ_ESTIMATED
42	Y	estimated	3	Y	R9	A8 LATE_ADJ_ESTIMATED
43	Y	missing	1	N	R11	A10 CLOSED_STANDS_WATCH
44	Y	missing	1	Y	R11	A10 CLOSED_STANDS_WATCH
45	Y	missing	2	N	R11	A10 CLOSED_STANDS_WATCH
46	Y	missing	2	Y	R11	A10 CLOSED_STANDS_WATCH
47	Y	missing	3	N	R12	A10 CLOSED_STANDS_WATCH
48	Y	missing	3	Y	R10	A9 LATE_ADJ_DECLARED

6 Derived test set: decision table

Each condensed rule has one **rule test**. Each don't-care has **invariance tests** that prove the ignored condition really is ignored. The complete 48-row table is also run as one parameterised test (TC-DT-ALL).

6.1 Rule tests (one per rule)

ID	Rule	Input (source, scope, decl, closed)	Expected outcome	Req
TC-DT-01	R1	metered, 2, N, N	A1. In period total; tier METERED; U = 2% of item.	FR-UNC-01
TC-DT-02	R2	invoiced, 1, N, N	A2. In period total; tier INVOICED; U = 8%.	FR-UNC-01
TC-DT-03	R3	estimated, 3, N, N	A3. In period total; tier ESTIMATED; U = 25%; item listed as estimated.	FR-UNC-06
TC-DT-04	R4	missing, 3, Y, N	A4. Declared value used; tier ESTIMATED; flag DECLARED; evidence hash stored.	FR-IN-06
TC-DT-05	R5	missing, 1, N, N	A5. Open Item raised; no value in totals; close refused while unresolved; at close gap-filled from latest earlier month.	FR-IN-05, FR-CALC-06
TC-DT-06	R6	missing, 3, N, N	A5. As TC-DT-05 for Scope 3 without declaration.	FR-IN-05
TC-DT-07	R7	metered, 2, N, Y	A6. Stored with is_late; closed report hash unchanged; adjustment queued for next open period.	FR-LATE-01 to 05
TC-DT-08	R8	invoiced, 1, N, Y	A7. As TC-DT-07 with tier INVOICED.	FR-LATE-01 to 05
TC-DT-09	R9	estimated, 3, N, Y	A8. As TC-DT-07 with tier ESTIMATED.	FR-LATE-01 to 05
TC-DT-10	R10	missing, 3, Y, Y	A9. Late declaration is a late record; adjustment = declared emission minus gap-filled amount in the closed report; tier ESTIMATED; flag DECLARED.	FR-LATE-09

6. Derived test set: decision table

ID	Rule	Input (source, scope, decl, closed)	Expected outcome	Req
TC-DT-11	R11	missing, 1, N, Y	A10. Closed report unchanged; item on watchlist; no adjustment posted.	FR-LATE-08
TC-DT-12	R12	missing, 3, N, Y	A10. As TC-DT-11 for Scope 3.	FR-LATE-08

6.2 Invariance tests (don't-cares)

ID	Combination	Expected	Proves
TC-DT-13	invoiced, scope 1, Y, open	Same as TC-DT-02 (A2); declaration stored as evidence only.	Declaration does not change a Scope 1 outcome.
TC-DT-14	metered, scope 2, Y, open	Same as TC-DT-01 (A1).	Same for Scope 2.
TC-DT-15	estimated, scope 3, Y, open	Same as TC-DT-03 (A3); tier stays ESTIMATED, not upgraded by the declaration.	Declaration never upgrades a tier.
TC-DT-16	missing, scope 1, Y, open	A5 (not A4): Open Item raised; declaration ignored.	The awkward case: declaration present, but Scope 1.
TC-DT-17	missing, scope 2, Y, closed	A10 (not A9).	Declaration ignored for Scope 2 after close.
TC-DT-18	invoiced, scope 3, Y, closed	Same as TC-DT-08 (A7).	Late-path tier is that of the record.
TC-DT-ALL	all 48 rows	For each row the system output equals the table (rule, action, tier, included-in-total flag, effect on closed report).	Complete coverage; NFR-MNT-01.

6.3 Awkward combinations and boundaries

ID	Scenario	Expected result	Req
TC-AW-01	Closed, missing, Scope 1, declaration Y.	A10; declaration ignored; no adjustment.	FR-LATE-08
TC-AW-02	Closed, missing, Scope 3, declaration Y arrives.	A9 with delta against the gap-filled amount (fixture FX-04).	FR-LATE-09
TC-AW-03	Closed, missing, Scope 3, no declaration.	A10 stands; watchlist reminder generated.	FR-LATE-08
TC-AW-04	Open, missing, Scope 3, declaration Y.	A4: declared value used as ESTIMATED, flag DECLARED.	FR-IN-06
TC-AW-05	Open, missing, no declaration, and no earlier month exists (first month of operation).	Close refused with a message naming the item; after a record is entered, close succeeds.	FR-CALC-06
TC-AW-06	Late estimated record replaces a gap-filled item (FX-03: gap-fill 3,000 L = 7,500 kg; actual 3,400 L).	Adjustment = 8,500 - 7,500 = +1,000 kg; tier ESTIMATED. A second case with actual 2,400 L (6,000 kg) gives -1,500 kg.	FR-LATE-04
TC-AW-07	Late metered reading corrects an invoiced figure.	Adjustment tier is METERED ($\pm 2\%$); delta uses the amount most recently reported.	FR-LATE-04
TC-AW-08	Activity date 2024-12-31 and 2025-01-01, same activity type (fixture factors 2.5 and 2.4).	2024-12-31 uses 2.500000; 2025-01-01 uses 2.400000 (half-open interval).	FR-FAC-03

6. Derived test set: decision table

ID	Scenario	Expected result	Req
TC-AW-09	Record stored at exactly <code>closed_at</code> , and one at <code>closed_at</code> + 1 second.	First is in the closed report; second is late.	FR-PC-05
TC-AW-10	Two successive corrections of the same item after close.	Second adjustment is against the first adjustment's amount, not the original; no double counting.	FR-LATE-04
TC-AW-11	Electricity bill 2024-12-16 to 2025-01-15, 31,000 kWh (1,000 kWh a day).	Split 16,000 kWh to 2024 ($\times 0.500000 = 8,000$) and 15,000 kWh to 2025 ($\times 0.450000 = 6,750$).	FR-IN-04
TC-AW-12	A record of 0 kWh with evidence, versus no record.	First is valid and counted as zero; second is an Open Item.	FR-IN-07

7 Other functional tests

7.1 Intake and calculation

ID	Test	Expected	Req
TC-IN-01	Create a record with all fields.	Stored; <code>recorded_at</code> set by the system, not by the user.	FR-IN-01
TC-IN-02	Attempt UPDATE and DELETE on an activity record through SQL and the API.	Both rejected; correction by superseding record works.	FR-IN-02
TC-IN-03	Try to upgrade estimated to invoiced without evidence, then with evidence.	Refused, then accepted; downgrade always accepted.	FR-IN-03
TC-CA-01	FX-02 record r1: 100,000 kWh $\times$ 0.500000.	50,000.000000 kgCO ₂ e.	FR-CALC-01
TC-CA-02	Quantity $3 \times$ factor 0.3333335 (tie at the 7th decimal).	ROUND_HALF_EVEN result identical on every run and platform.	FR-CALC-02
TC-CA-03	Record in MWh with factor per kWh.	Conversion taken from the reference release; changing a code constant has no effect.	FR-CALC-03
TC-CA-04	FX-02 full set.	Scope 1 = 10,000; Scope 2 = 50,000; Scope 3 = 2,000; total 62,000 (all 6 d.p.).	FR-CALC-04
TC-CA-05	Supplier declares 1,250 kgCO ₂ e directly.	Used as given; marked "no factor applied"; tier ESTIMATED.	FR-CALC-05

7.2 Uncertainty (figures from the project: 2%, 8%, 25%)

Fixture FX-02 emissions: r1 = 50,000 metered, r2 = 10,000 invoiced, r3 = 2,000 estimated (kgCO₂e).

ID	Test	Expected	Req
TC-UN-01	Item uncertainties for FX-02.	$U(r1) = 1,000$; $U(r2) = 800$; $U(r3) = 500$ kgCO ₂ e.	FR-UNC-01, 02
TC-UN-02	RSS of FX-02.	$\sqrt{1,000,000 + 640,000 + 250,000} = \mathbf{1,374.772708}$ kgCO ₂ e.	FR-UNC-03
TC-UN-03	Worst-case bound of FX-02.	$1,000 + 800 + 500 = \mathbf{2,300.000000}$ kgCO ₂ e, shown next to the RSS value.	FR-UNC-04

7. Other functional tests

ID	Test	Expected	Req
TC-UN-04	Disclosure format. FX-02: total 62,000. Second fixture: items 1,000 metered, 2,000 invoiced, 500 estimated.	FX-02: 62,000 ± 1,400 kgCO₂e (2.2%) . Second: U(i) = 20, 160, 125; RSS 204.022058; total 3,500; shown as 3,500 ± 200 kgCO₂e (5.8%) ; worst-case 305.	FR-UNC-05
TC-UN-05	Tier shares for FX-02.	Metered 80.6%, invoiced 16.1%, estimated 3.2% of total emissions; gap-filled and declared lists present (empty here).	FR-UNC-06
TC-UN-06	Scan screens, PDF and CSV of a report.	Every total is followed by its uncertainty; a total without it fails the test.	FR-UNC-07
TC-UN-07	Method statement.	Present and complete in every report: 2%, 8%, 25%; RSS; independence assumption; worst-case bound; factor uncertainty excluded.	FR-UNC-08
TC-UN-08	Ten identical estimated items of 100 kgCO ₂ e.	RSS = $\sqrt{10 \times 25 \times 25} = 79.056942$; worst-case = 250. Shows why both are disclosed.	FR-UNC-03, 04

7.3 Emission factor versioning

Synthetic factors for tests only (not authority values):

Activity type	Unit	2024 value	2025 value
ELEC_GRID	kWh	0.500000	0.450000
FUEL_DIESEL	litre	2.500000	2.400000
FREIGHT_ROAD	tonne-km	0.100000	0.090000

Releases: **RR-T2024.1** (2024 rows), **RR-T2025.1** (2024 and 2025 rows), **RR-T2026.1** (as RR-T2025.1, but the 2024 diesel factor corrected retroactively to 2.550000).

ID	Test	Expected	Req
TC-FV-01	UPDATE, DELETE and INSERT on a sealed release.	All rejected by the database; audit log shows the attempt.	FR-FAC-01
TC-FV-02	Inspect a loaded row.	Has authority, edition, type, unit, value, valid_from, valid_to, checksums, loaded_at.	FR-FAC-02
TC-FV-03	Look up diesel for 2024-06-15 in RR-T2025.1; look up a type with no row.	2.500000; second gives a hard error, no fallback to another year or release.	FR-FAC-03
TC-FV-04	Insert an overlapping interval in an unsealed release.	Rejected by the exclusion constraint.	FR-FAC-04
TC-FV-05	Change one byte of a factor file, then load or use the release.	Manifest hash mismatch; load and use refused.	FR-FAC-05
TC-FV-06	Run what-if with RR-T2026.1 on the 2024 period.	Output stamped COMPARISON; cannot be issued; stored report unchanged.	FR-FAC-07
TC-FV-07	Load RR-T2026.1 after R-T2024-A was issued.	New release created; R-T2024-A still pinned to RR-T2024.1; its hash unchanged.	FR-FAC-08

7. Other functional tests

7.4 Period close

ID	Test	Expected	Req
TC-PC-01	Walk a period through its states.	OPEN, CLOSING, CLOSED only; illegal transitions refused.	FR-PC-01
TC-PC-02	Attempt close with (a) an unresolved Open Item, (b) a metered record without evidence, (c) a factor lookup error, (d) a release hash mismatch.	Each refused with a specific reason; period stays OPEN.	FR-PC-02
TC-PC-03	Close as Reporting Lead, then as another role.	First succeeds and stores report, hash and <code>closed_at</code> ; second refused.	FR-PC-03
TC-PC-04	Enter a record between the data cut-off and close.	Accepted into the open period, flag AFTER_CUTOFF; not treated as late.	FR-PC-04
TC-PC-05	Add records after close; recompute with cut-off = <code>closed_at</code> .	Only records with <code>recorded_at</code> ≤ <code>closed_at</code> are used.	FR-PC-05
TC-PC-06	Try to reopen a closed period.	Refused; only restatement is offered.	FR-PC-06

7.5 Late invoice after close

Fixture FX-04. Period FY2024 closed at 2025-01-15T10:00:00Z as report R-T2024-A. Late invoice r4: Plant P1, Scope 1, 1,000 litres diesel, invoiced, activity date 2024-12-20, stored 2025-01-20.

ID	Test	Expected	Req
TC-LT-01	Store r4.	Accepted; <code>is_late</code> true; attributed = FY2024; posted = earliest open period; Reporting Lead notified.	FR-LATE-01, 07
TC-LT-02	Compare R-T2024-A hash before and after storing r4; run verify.	Identical hash; verify PASS.	FR-LATE-02
TC-LT-03	Check the factor used for r4, with RR-T2026.1 (2.550000) and RR-T2025.1 (2.400000) loaded.	2.500000 (valid on 2024-12-20, from the release pinned by R-T2024-A). Emission 2,500.000000; U = 200.000000 (8%).	FR-LATE-03
TC-LT-04	Late correction r2b supersedes r2 (4,000 L reported as 10,000 kg): (a) 4,200 L, (b) 3,800 L.	(a) +500.000000; (b) -500.000000.	FR-LATE-04
TC-LT-05	Generate the next period's report.	Separate "prior-period adjustments" section with r4 (attributed FY2024); subtotals "this period" and "this period plus adjustments".	FR-LATE-05
TC-LT-06	Reporting Lead restates FY2024 (criterion met by configuration).	New version R-T2024-A v2 with new hash; v1 SUPERSEDED and still verifies PASS; r4 marked ABSORBED and absent from the next period's report.	FR-LATE-06
TC-LT-07	A plant stores an invoice for a period still OPEN.	Normal path; no adjustment line.	FR-LATE-01

7. Other functional tests

7.6 Audit, security and performance

ID	Test	Expected	Req
TC-AU-01	Ingest, close, restate, load release, verify.	One audit entry each, with actor, time and hashes; entries cannot be edited.	FR-AUD-01
TC-AU-02	Drill from a total to line, record, factor row, evidence file.	Every hop resolves; evidence hash matches the file.	FR-AUD-02
TC-AU-03	Role matrix: each role tries each action.	Only permitted actions succeed; Plant Data Officer cannot see other plants.	FR-AC-01, NFR-SEC-01
TC-PF-01	Recompute a full period with nine plants of generated data.	Completes within the 10 minute target.	NFR-PERF-01

8

Reproducibility regression test

Requirement tested: NFR-REP-01, FR-RPT-02, FR-RPT-04, FR-PC-05, FR-FAC-06, FR-FAC-07. A report issued in 2024 recomputes **byte-identically** in 2026.

8.1 Golden artefacts (stored in Git under tests/golden/R-T2024-A/)

File	Content
pair.lock	Pair ID, calc module, git commit, reference release id and manifest hash, lock-file hash, container digest.
inputs.csv	The activity records as of <code>closed_at</code> , in canonical order, with their <code>recorded_at</code> .
inputs.sha256	SHA-256 of <code>inputs.csv</code> .
report.canonical.json	The report as issued (FR-RPT-02 format).
report.sha256	SHA-256 of the report bytes, as printed on the issued PDF.
closed_at.txt	2025-01-15T10:00:00Z.

8.2 Procedure of TC-RP-04 (the golden test)

1. Start a clean, empty test database in the pinned container.
2. Load the reference releases named by `pair.lock` and verify their manifest hashes (FR-FAC-05).
3. Load `inputs.csv`; check its SHA-256 equals `inputs.sha256` (stops here if inputs differ).
4. Run `verify` with the pair's calc module and `cut-off = closed_at`.
5. Compare the produced canonical JSON with `report.canonical.json` as **bytes**. Pass only if equal and the SHA-256 equals `report.sha256`.

```
def test_golden_report_recomputes_byte_for_byte(clean_db):
    g = Path("tests/golden/R-T2024-A")
    pair = load_pair_lock(g / "pair.lock")
    clean_db.load_release(pair.refdata_release)           # hash verified inside
```

8. Reproducibility regression test

```
raw = (g / "inputs.csv").read_bytes()
assert sha256(raw).hexdigest() == (g / "inputs.sha256").read_text().strip()
clean_db.load_inputs(raw)
produced = recompute(pair, cutoff=read_instant(g / "closed_at.txt"), db=clean_db)
expected = (g / "report.canonical.json").read_bytes()
assert produced == expected # byte for byte
assert sha256(produced).hexdigest() == (g / "report.sha256").read_text().strip()
```

8.3 The test set around it

ID	Test	Expected	Req
TC-RP-01	Issue a report.	Stored with Pair ID, calc module and reference release id.	FR-FAC-06
TC-RP-02	Inspect PDF and CSV renderings.	All FR-RPT-01 sections present; report hash printed.	FR-RPT-01, 03
TC-RP-03	Run the same report twice, with a different time zone, locale, PYTHONHASHSEED , and shuffled input row order.	Four identical byte outputs.	FR-RPT-02, NFR-REP-03
TC-RP-04	Golden test of 8.2.	Bytes equal; hashes equal.	NFR-REP-01
TC-RP-05	Two years later simulation. Freeze the clock at 2026-10-03; load RR-T2025.1 and RR-T2026.1 (diesel 2024 corrected to 2.550000); add a calc.v2 module with a different rounding rule; store late record r4 and correction r2b; then run TC-RP-04.	Still byte-identical to the golden file. Newer factors, newer code and later data do not leak in.	NFR-REP-01
TC-RP-06	Mutation check. In a scratch copy: change diesel 2024 to 2.500001; or change one input quantity; or change the rounding mode.	In each case the produced bytes differ and verify returns FAIL. Proves the golden test can detect change.	FR-RPT-04
TC-RP-07	Remove the pinned release, or alter its manifest. Run verify .	FAIL with an explicit reason; the system never substitutes another release.	FR-FAC-07
TC-RP-08	Alter one byte of inputs.csv .	Input hash check fails before any calculation.	FR-RPT-04

8.4 When it runs

- On **every commit** in CI: all golden directories (one per issued report), so any change that alters an old number is caught before merge.
- On **every release tag**: the full suite, plus TC-RP-05 and TC-RP-06.
- **Once a year** as a reproduction drill: restore last year's archive onto a fresh machine and run **verify** on every issued report (CL-CMP-001, section 9).
- A new golden directory is added **every time a report is issued** (part of the issue checklist).

9. Traceability matrix

Requirement	Tests
FR-IN-01 to 03	TC-IN-01, TC-IN-02, TC-IN-03
FR-IN-04	TC-AW-11
FR-IN-05, FR-IN-06, FR-IN-07	TC-DT-05, TC-DT-06, TC-DT-04, TC-AW-04, TC-AW-12
FR-CALC-01 to 05	TC-CA-01 to TC-CA-05
FR-CALC-06	TC-DT-05, TC-AW-05
FR-FAC-01 to 08	TC-FV-01 to TC-FV-07, TC-AW-08, TC-RP-01, TC-RP-07
FR-UNC-01 to 08	TC-DT-01 to 03, TC-UN-01 to TC-UN-08
FR-PC-01 to 06	TC-PC-01 to TC-PC-06, TC-AW-09
FR-LATE-01 to 09	TC-LT-01 to TC-LT-07, TC-DT-07 to TC-DT-12, TC-AW-01 to 03, TC-AW-06, 07, 10
FR-RPT-01 to 04	TC-RP-02, TC-RP-03, TC-RP-04, TC-RP-08
FR-AUD-01, 02, FR-AC-01	TC-AU-01 to TC-AU-03
NFR-REP-01 to 03	TC-RP-03 to TC-RP-07
NFR-PERF-01, NFR-SEC-01	TC-PF-01, TC-AU-03
NFR-MNT-01	TC-DT-ALL

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Test environment, entry and exit criteria

Environment: the pinned container image (Python 3.12, PostgreSQL 16, dependencies fixed by a hash-checked lock file); clean database per test run; clock injected, never read from the machine; time zone UTC.

Entry criteria: requirements baseline (SRS v1.0) approved; fixtures FX-01 to FX-04 and the golden directory created; CI green on unit tests.

Exit criteria: all test cases pass; all 12 rules and all 48 combinations exercised; TC-RP-04 and TC-RP-05 pass and TC-RP-06 fails the mutated runs as expected; no open defect that changes a reported number; results recorded against the Pair that was tested.

Defect handling: a defect that changes any reported number is never fixed by editing a frozen calc module. It is fixed in a new calc module and reaches old periods only through an explicit restatement (see CL-CMP-001, section 3.4).